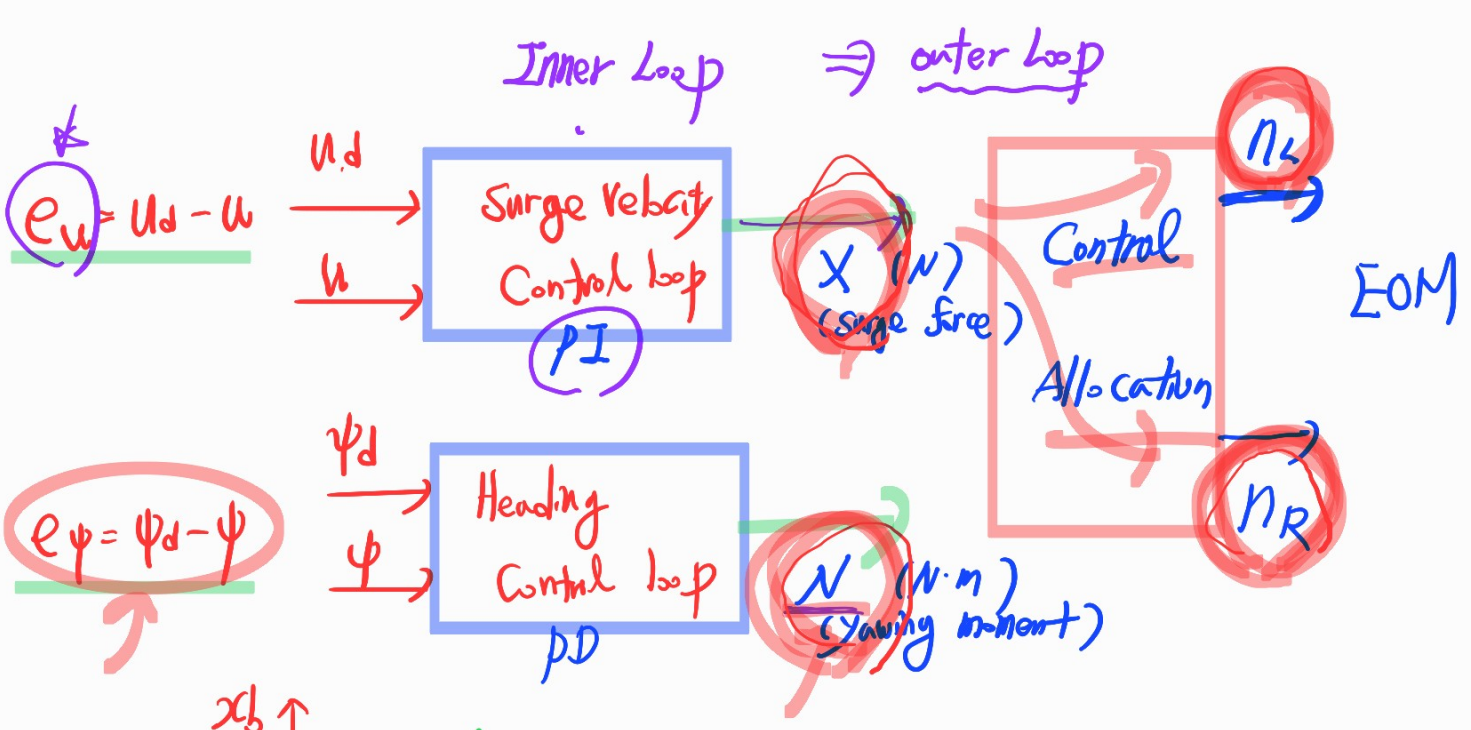
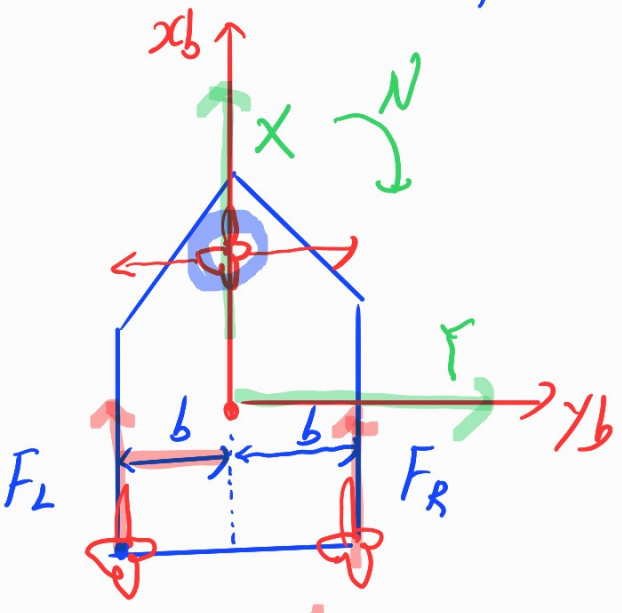




$$\begin{bmatrix} X \\ Y \\ N \end{bmatrix} = \begin{bmatrix} F_L + F_R \\ 0 \\ b(F_L - F_R) \end{bmatrix}$$

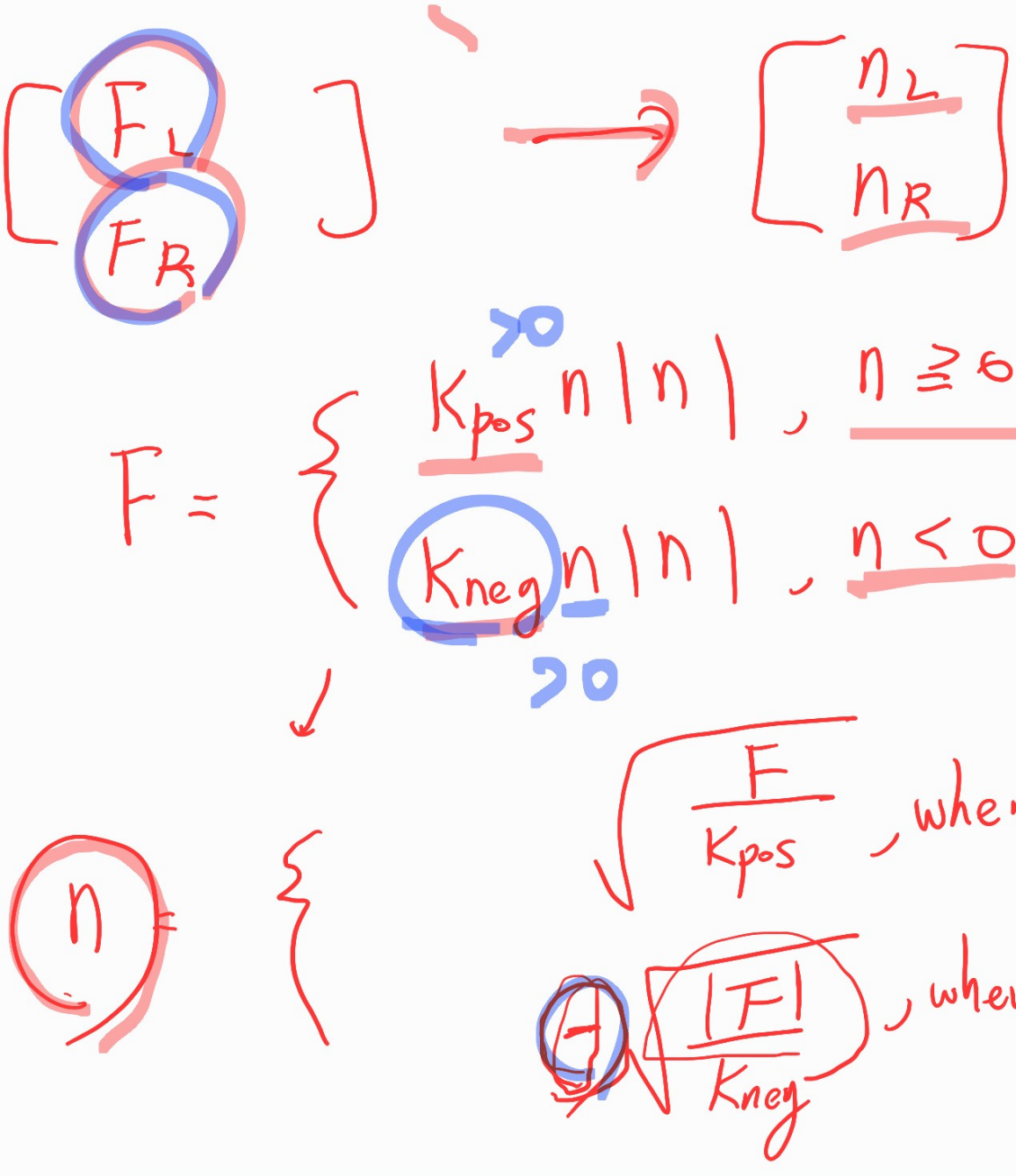
$$\begin{bmatrix} X \\ N \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ b & -b \end{bmatrix} \begin{bmatrix} F_L \\ F_R \end{bmatrix}$$

T

Controller output

$$\begin{bmatrix} F_L \\ F_R \end{bmatrix} = T^{-1} \cdot \begin{bmatrix} X \\ N \end{bmatrix}$$

Known


F_L
F_R
η_L
η_R
$F = \begin{cases} K_{pos}\,\eta|\eta|, & \eta \geq 0 \\ K_{neg}\,\eta|\eta|, & \eta < 0 \end{cases}$
$K_{pos} > 0$
$K_{neg} > 0$
$\eta = \begin{cases} \sqrt{\frac{F}{K_{pos}}}, & \text{when } F \geq 0 \\ -\sqrt{\frac{|F|}{K_{neg}}}, & \text{when } F < 0 \end{cases}$

Heading Control : PD

Yaw rate feedback

$$N = K_p e_\psi - K_d \cdot r$$

where $e_\psi = \psi_d - \psi(x)$

$-180° \sim 180°$

$\frac{de(t)}{dt}$

$$+ K_d \cdot \frac{N \cdot s}{s+N} \cdot e_\psi$$

pseudo-derivate

$$\therefore N = K_p \cdot ssa(e_\psi) - K_d \cdot r$$

↳ to reduce %OS

wrap to $-\pi$ to π